



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid-term Exam, Trimester: Spring 2026

Course Code: EEE-2123 Course Title: Electronics

Total Marks: 30 Duration: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. I/V characteristics of a diode operating at a fixed temperature is given in Fig 1. (for $n = 1.x$; where x is your last digit of your ID. For example, If the last digit of your ID is 2, then $n=1.2$)

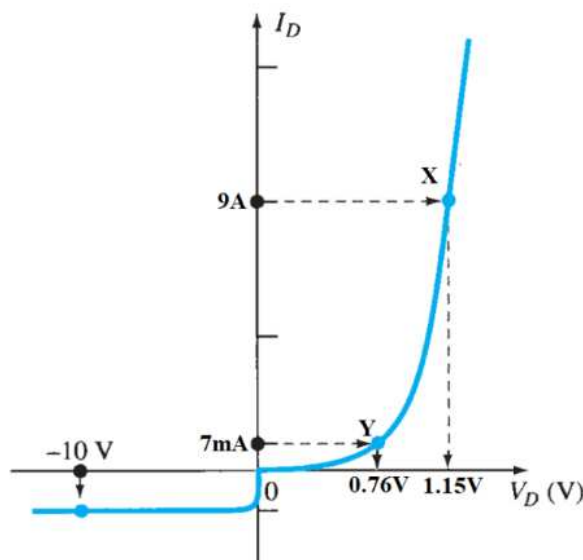


Fig 1: Figure for question 1

Determine the followings:

- (a) The operating temperature of the diode. [Hint: Use $I_D = I_S e^{V_D/nV_T}$] [2]
- (b) Reverse saturation current at the operating temperature. [2]
- (c) Find static resistance for $1.15V$ for a different operating temperature of $80^\circ C$ with reference to the given I/V curve. [2]

2. Determine V and I from the circuit shown in Fig 2. Assume that both of the diodes are ideal. [6]

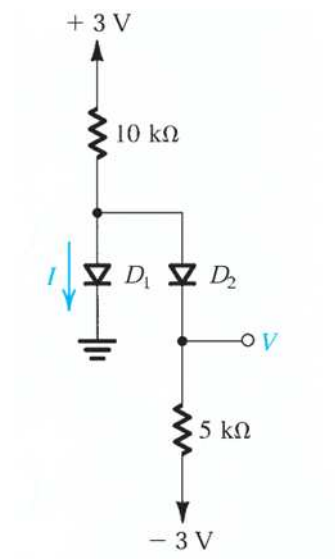


Fig 2: Figure for question 2.

3. In Fig 3, the input is a sine wave with 75 V amplitude. The voltage across the $R5$ resistor is the output (which is denoted by V_o). Assume the diodes have a breakdown voltage of 40V. Answer the followings:

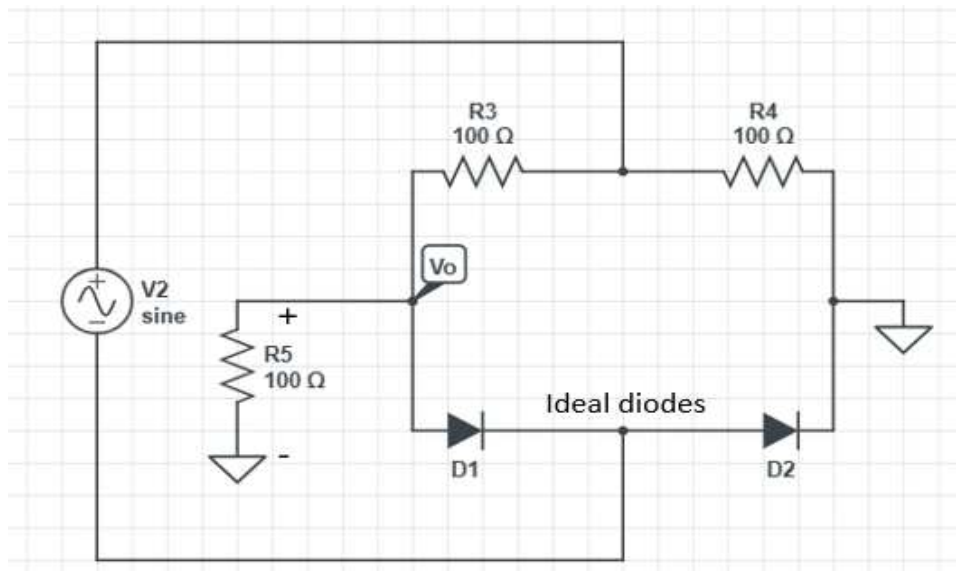


Fig 3: Figure for question 3

- Determine** the exact relationship between input and output and **sketch** the output with proper voltage labeling. [3]
- What would you comment on the **safety** of this circuit? **Explain** with proper calculations if necessary. [3]

4. In the circuit of Fig 4, the input is a sine wave of 10V amplitude. **Sketch** the waveforms of I_x and I_y with peak values properly labeled. [3 + 3]

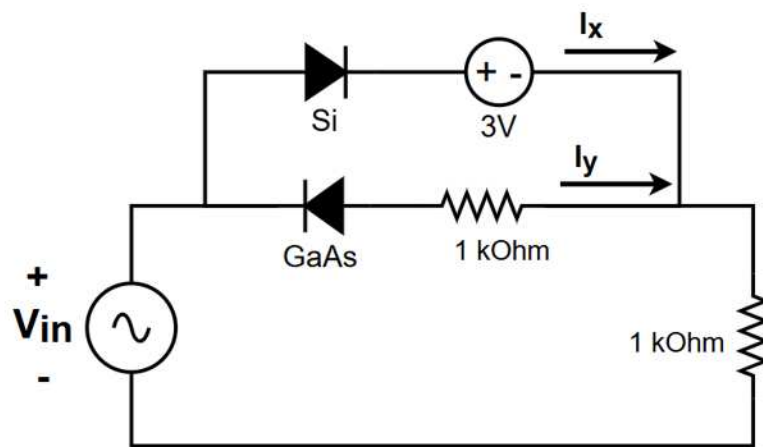


Fig 4: Figure for question 4

5. Look carefully at the diagram in Fig 5. The input is consecutively passed through Circuit 1 and Circuit 2 and the output is obtained. Your task is to **design both the circuits**. In your design use **Si** diodes. [3+3]

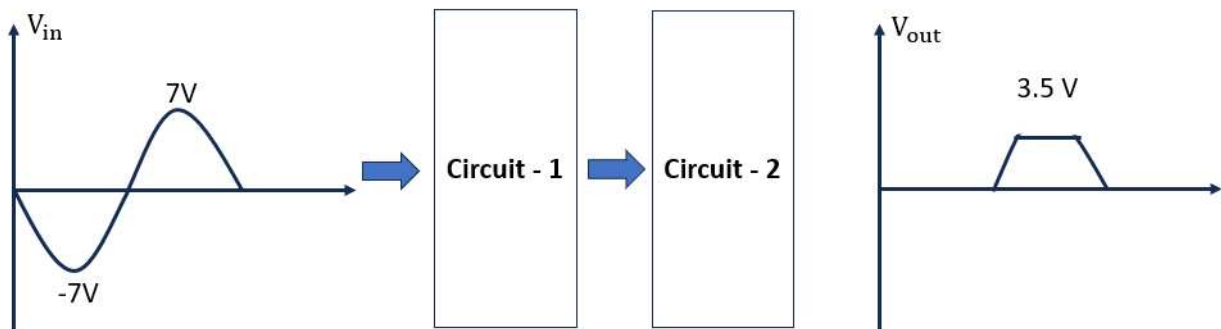


Fig 5: Figure for question 5